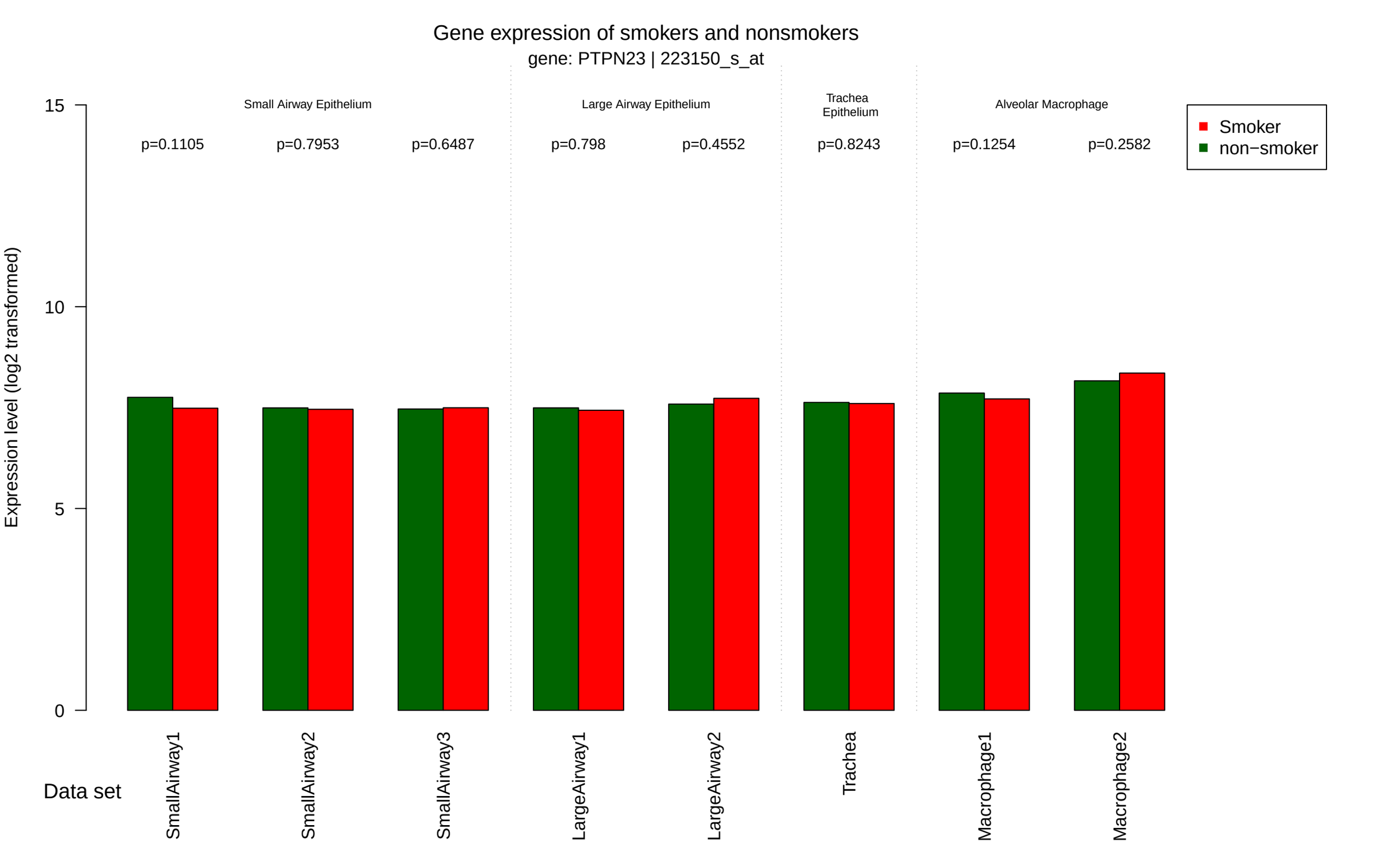
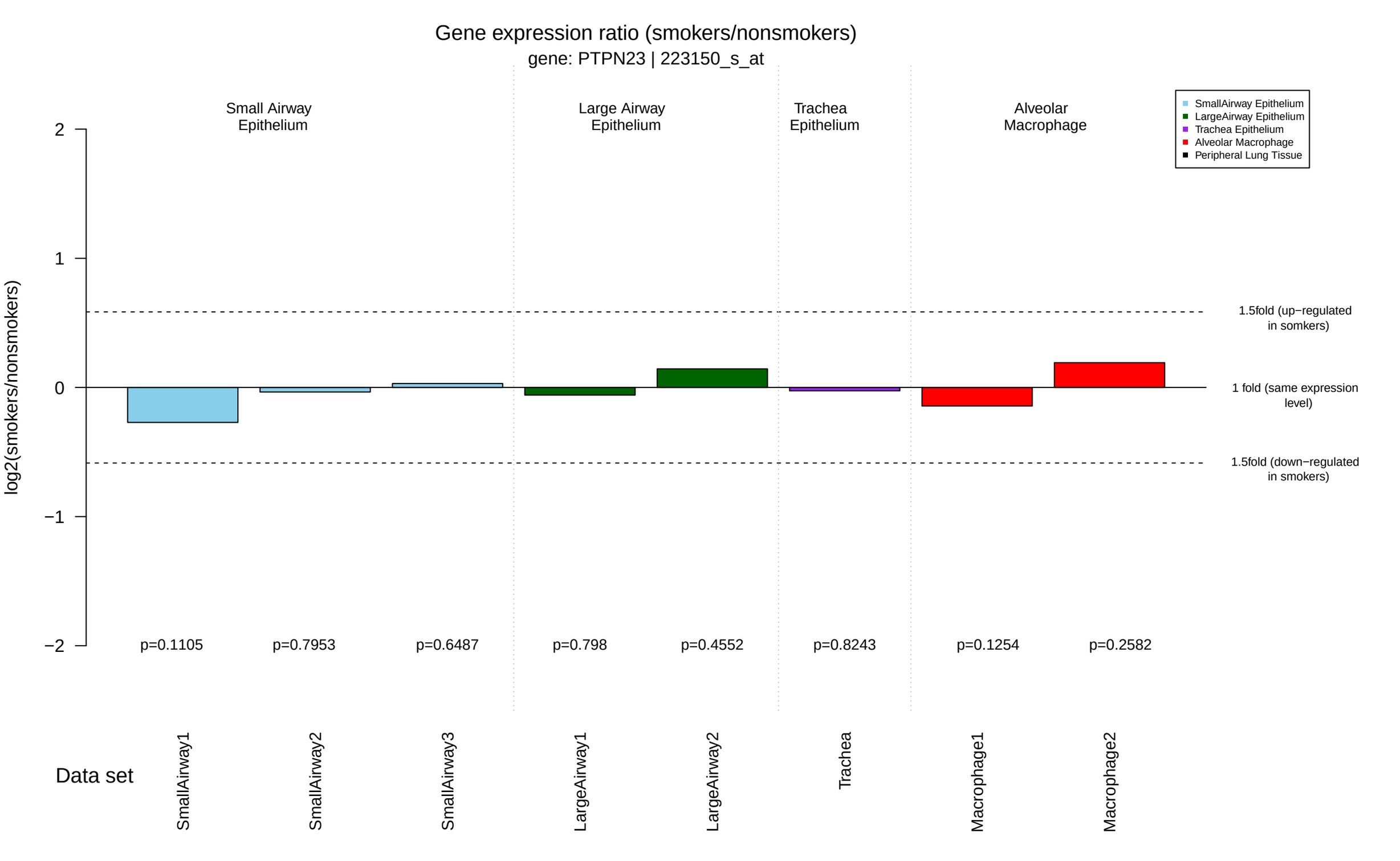
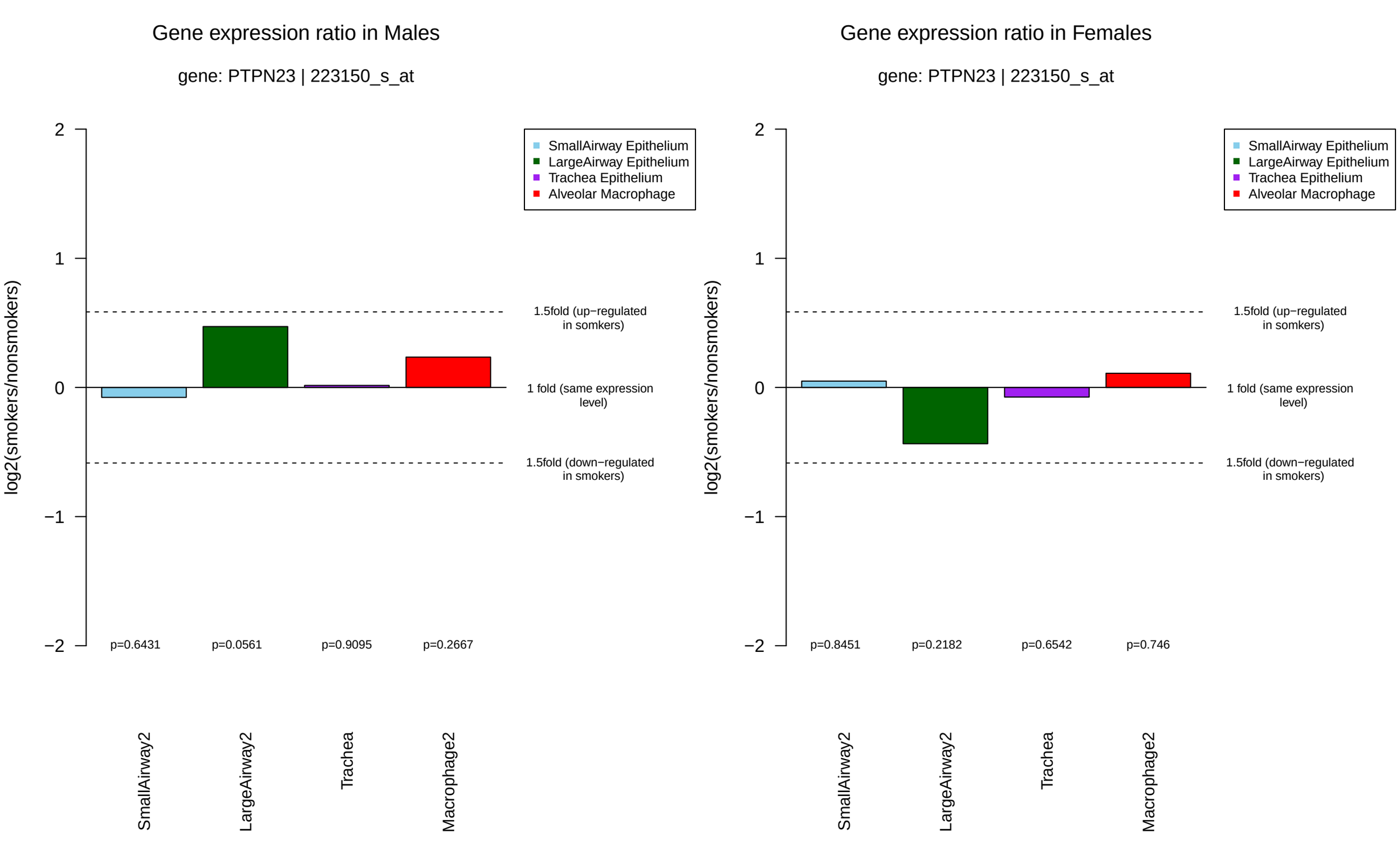
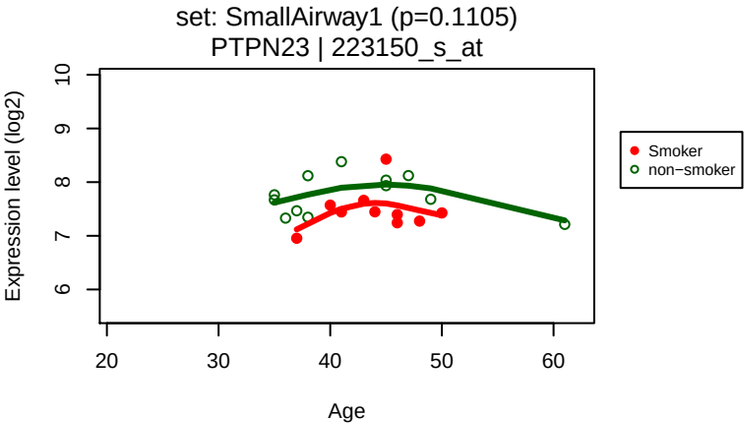
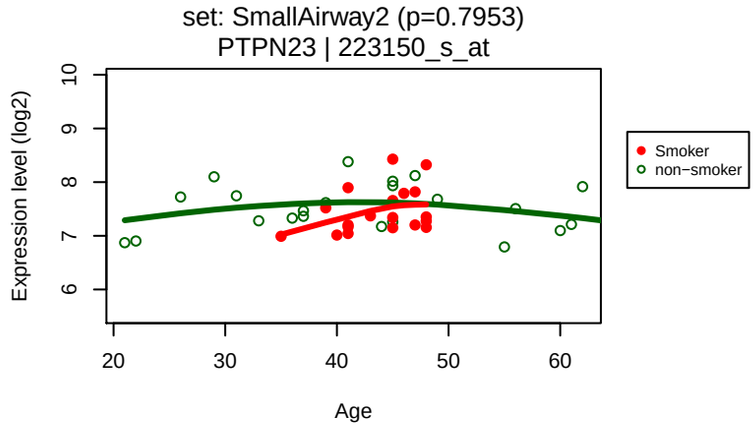
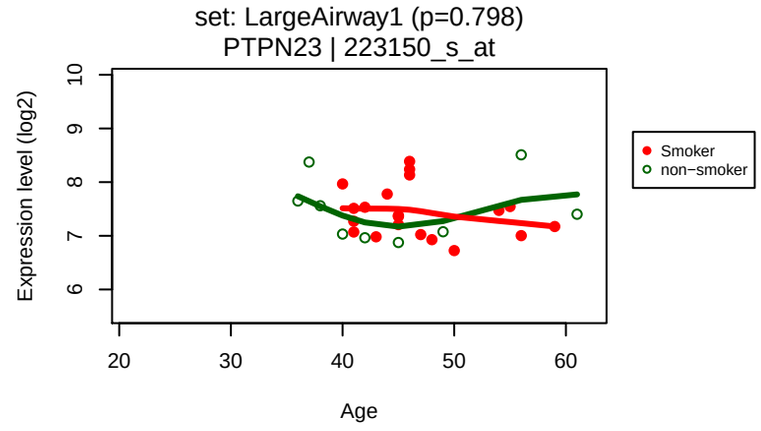

Gene expression of smokers and nonsmokers
gene: PTPN23 | 223150_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1105
p=0.7953
p=0.6487
p=0.798
p=0.4552
p=0.8243
p=0.1254
p=0.2582
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

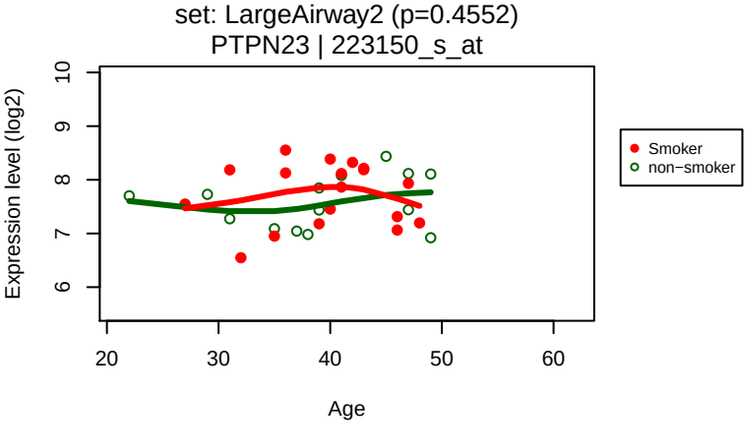

Gene expression ratio (smokers/nonsmokers)
gene: PTPN23 | 223150_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1105
p=0.7953
p=0.6487
p=0.798
p=0.4552
p=0.8243
p=0.1254
p=0.2582
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

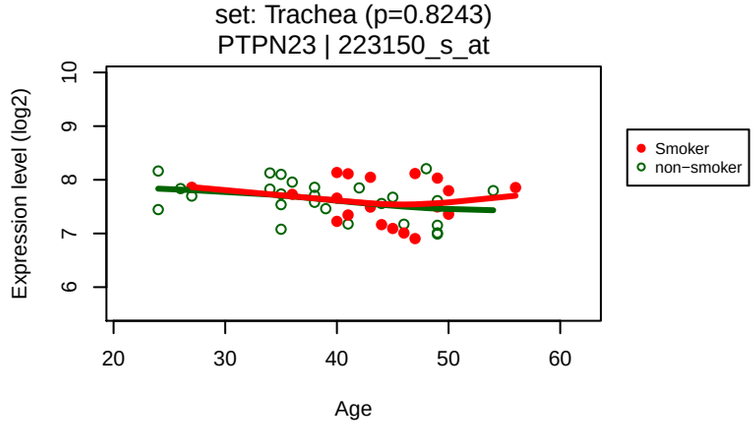

Gene expression ratio in Males
gene: PTPN23 | 223150_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6431
p=0.0561
p=0.9095
p=0.2667
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN23 | 223150_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8451
p=0.2182
p=0.6542
p=0.746
SmallAirway2
LargeAirway2
Trachea
Macrophage2

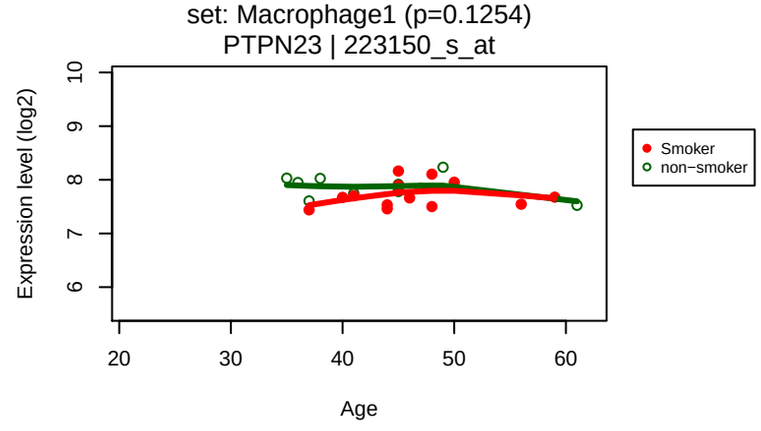

set: SmallAirway1 (p=0.1105)
PTPN23 | 223150_s_at
Expression level (log2)
Age
Smoker
non−smoker

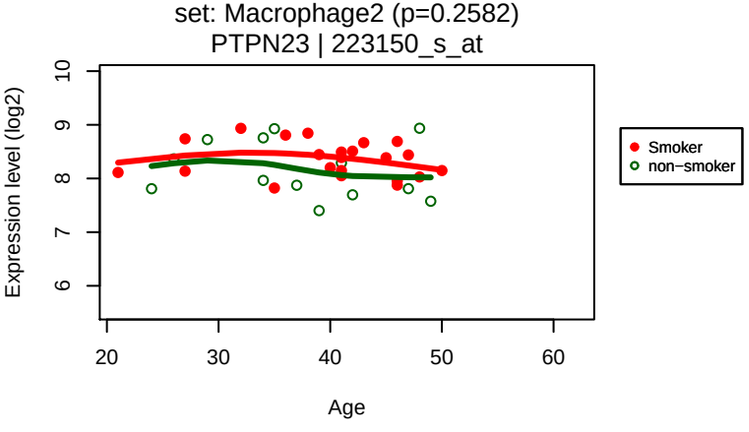

set: SmallAirway2 (p=0.7953)
PTPN23 | 223150_s_at
Expression level (log2)
Age
Smoker
non−smoker

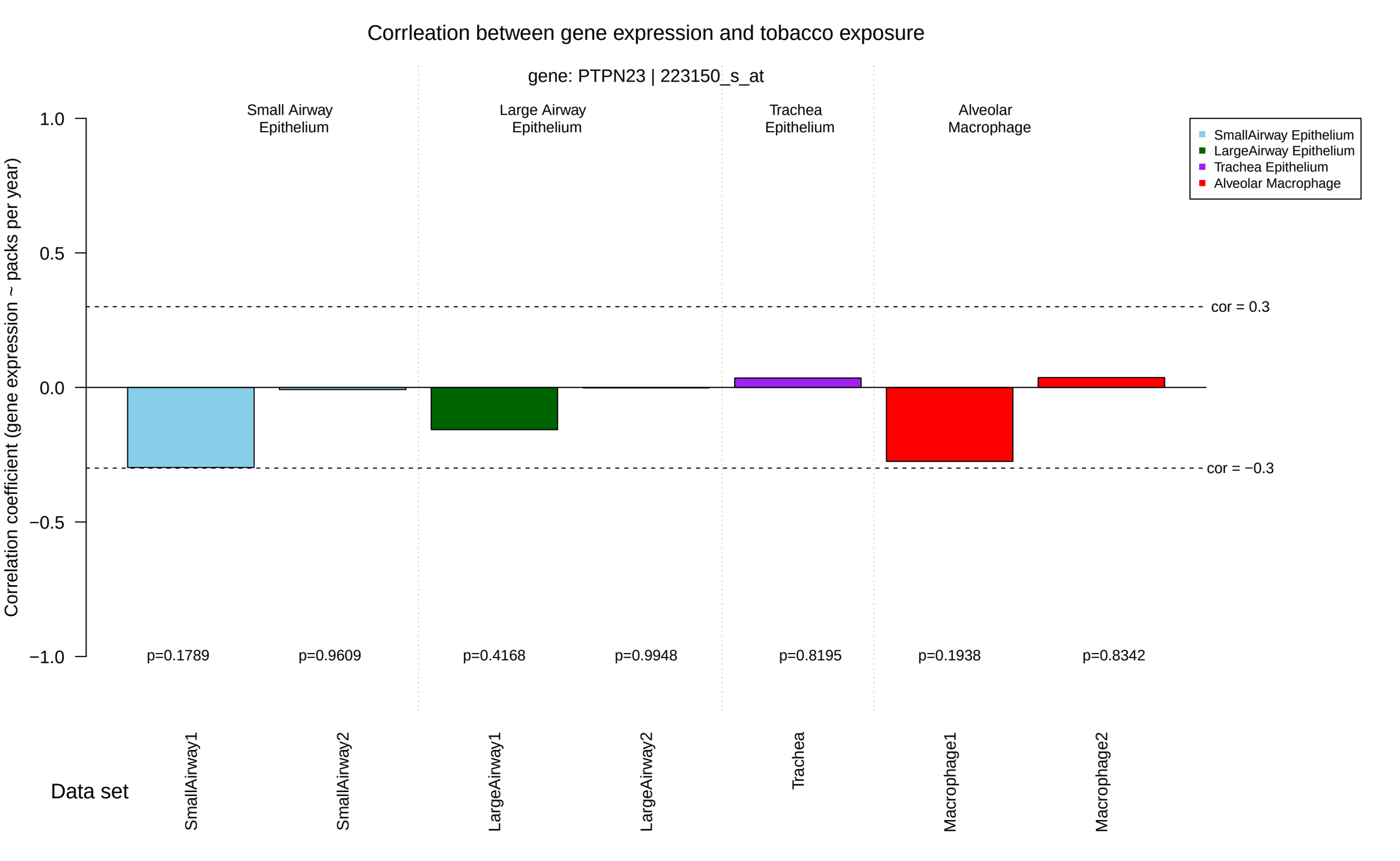

set: LargeAirway1 (p=0.798)
PTPN23 | 223150_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4552)
PTPN23 | 223150_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8243)
PTPN23 | 223150_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1254)
PTPN23 | 223150_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2582)
PTPN23 | 223150_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN23 | 223150_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1789
p=0.9609
p=0.4168
p=0.9948
p=0.8195
p=0.1938
p=0.8342
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2